

## Article

# DEEPEIA: Conceptualizing a Generative Deep Learning Foreign Market Recommender for SMEs

Nuno Calheiros-Lobo <sup>1,\*</sup> , Manuel Au-Yong-Oliveira <sup>1,2</sup>  and José Vasconcelos Ferreira <sup>1</sup> 

<sup>1</sup> Research Unit on Governance, Competitiveness and Public Policies (GOVCOPP), Department of Economics, Management, Industrial Engineering and Tourism (DEGEIT), University of Aveiro, 3810-193 Aveiro, Portugal; mao@ua.pt (M.A.-Y.-O.); josev@ua.pt (J.V.F.)

<sup>2</sup> Institute for Systems and Computer Engineering, Technology and Science (INESC TEC), 4200-465 Porto, Portugal

\* Correspondence: ncl@ua.pt; Tel.: +351-919-749-749

## Abstract

This study introduces the concept of DEEPEIA, a novel deep learning (DL) platform designed to recommend the optimal export market, and its ideal foreign champion, for any product or service offered by a small and medium-sized enterprise (SME). Drawing on expertise in SME internationalization and leveraging recent advances in generative artificial intelligence (AI), this research addresses key challenges faced by SMEs in global expansion. A systematic review of existing platforms was conducted to identify current gaps and inform the conceptualization of an advanced generative DL recommender system. The Discussion section proposes the conceptual framework for such a decision optimizer within the context of contemporary technological advancements and actionable insights. The conclusion outlines future research directions, practical implementation strategies, and expected obstacles. By mapping the current landscape and presenting an original forecasting tool, this work advances the field of AI-enabled SME internationalization while still acknowledging that more empirical validation remains a necessary next step.

**Keywords:** SME internationalization; artificial intelligence; deep learning; forecasting tool; conceptualization



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## 1. Introduction

AI is revolutionizing SME internationalization, offering unprecedented opportunities for global market expansion. This disruptive technology, characterized by its ability to learn, adapt, and make data-driven decisions [1,2], is reshaping how SMEs approach foreign markets.

### 1.1. AI and SME Internationalization

The unique capacity of AI to autonomously refine its algorithms and derive rules from data [3–5] positions it as a powerful tool to help SMEs navigate international trade idiosyncrasies. AI applications in SME internationalization span various critical areas, including comprehensive risk management in global supply chains, the optimization of cross-border logistics, the enhancement of international manufacturing processes, and the development of sustainable business models [6–9].

Implementing AI in SMEs offers multifaceted benefits for international expansion, including efficient resource allocation, reduced operational costs, and the promotion of

sustainable global economic growth [10]. AI-driven Business Intelligence and Analytics (BI&BA) significantly enhances SMEs' competitive advantage in foreign markets [11].

However, SMEs face considerable challenges in AI adoption, such as limited expertise, high implementation costs, data security concerns, and integration complexities [12,13]. Despite these hurdles, the potential benefits, particularly in quality management and early issue detection in international operations, remain significant [14].

### 1.2. Research Objectives

This research aims to explore a gap in the existing literature by conceptualizing a novel DL platform designed to assist SMEs in identifying optimal foreign markets. This study builds upon a comprehensive systematic literature review on internationalization success determinants [15] and insights from expert interviews [16] to provide a more nuanced understanding of AI's potential in SME internationalization.

To address the critical gap in understanding AI's role in SME internationalization and to guide the development of their novel DL platform, the authors formulated the following research questions:

(RQ1) What is the current state of the art in DL foreign market entry recommenders?

(RQ2) What limitations do these platforms reveal, and how can they be solved?

(RQ3) How can a DL platform be conceptualized to successfully assist SMEs in their foreign market selection process?

Guided by these research questions, the authors established the following objectives to direct their investigation:

(O1) Enumerate the existing state-of-the-art DL foreign market recommenders.

(O2) Summarize their known limitations and ways to improve them.

(O3) Conceptualize a generative DL platform that identifies and explains the optimal foreign market to any SME product or service and proposes the ideal foreign champion for it.

### 1.3. Structure

This article opens with an introduction and background that contextualizes the study within the broader scholarly discourse. This is followed by a comprehensive synthesis of pertinent studies, which serves to construct the theoretical framework underpinning the research. The methodology section provides a detailed account of the research design, data collection, and analytical procedures employed. Subsequently, the results section systematically presents the empirical findings. The discussion critically evaluates this study's theoretical contributions, practical implications, acknowledged limitations, and outlines avenues for future research. Finally, this article concludes by offering a tangible tool designed to translate theoretical insights into actionable solutions for practitioners.

This research contributes to the evolving field of AI in SME internationalization, potentially transforming how SMEs approach and succeed in global markets.

## 2. Literature Review

### 2.1. Generative AI, DL, and Its Applications in SME Internationalization

The recent body of research pinpoints the growing significance of generative AI in recommender systems and business applications, particularly for SMEs engaging in internationalization. Generative models, such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), have demonstrated superior performance compared to traditional AI techniques in recommender systems [17,18]. These advanced models excel in generating personalized content, effectively addressing persistent challenges such as

data sparsity and cold start problems [19,20]. However, the specific application of these models in the context of SME internationalization remains scarce.

The potential of generative AI extends beyond conventional recommendation tasks, spanning various sectors relevant to SME internationalization, including hospitality, marketing education, and brand management [21–23]. Studies comparing the effectiveness of AI models like ChatGPT-4.5, Claude3, or Grok2 with existing AI recommenders have revealed similar influences on consumer purchase journeys while noting the greater impact of generative AI models on trust and consideration set adoption [24].

Innovative approaches in this field include the use of AI generators for content creation and repurposing, as well as the employment of semantic IDs for generative retrieval [25,26].

In the financial sector, generative AI is being used to predict firms' risk management capabilities and stock performance [27], which can be valuable for SMEs navigating international markets. Yet the applicability of these financial AI models to SMEs operating in emerging markets or niche sectors also requires further investigation.

Despite these advancements, implementing generative AI faces significant challenges, especially for SMEs. Oldemeyer et al. [13] identify issues such as a lack of knowledge, costs, and inadequate infrastructure as key barriers to AI adoption among SMEs. Privacy concerns and potential job displacement also need to be carefully addressed [28,29].

There is a pressing need to develop robust frameworks for assessing the ethical implications and societal impact of these advanced AI systems, ensuring their responsible and beneficial integration across various domains and international contexts. The development of AI solutions tailored to the needs of internationalizing SMEs remains an important gap in the current literature.

Deep learning (DL) has emerged as a powerful tool for decision support systems (DSS) in various fields, offering significant potential for SMEs in their internationalization efforts. DL's ability to automatically extract spatio-temporal features and transfer knowledge across domains makes it particularly suitable for complex decision-making environments faced by internationalizing SMEs [30–32].

Advanced DL models excel at processing multi-modal data, including text, images, and user behavior, to create sophisticated customer profiles across different international markets. This capability allows SMEs to tailor their offerings more effectively and understand nuanced market dynamics [33]. Deep learning techniques can enhance supply chain management for internationalizing SMEs by predicting demand, optimizing inventory, and identifying potential disruptions across global networks [34].

Generative models like Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) are being used to address challenges such as data sparsity and cold start problems in new markets [35].

However, implementing DL in DSS for SME internationalization faces several challenges [36]. Some scholars [37] highlight limitations related to data availability, computational demands, and interpretability issues. These challenges are particularly acute for SMEs, which often have limited resources and scarce data. The choice between DL and traditional ML methods depends on factors such as data size, computational resources, and the specific problem domain.

## 2.2. Decision Support Systems and Generative AI

Decision support systems (DSSs) have been applied to various domains, including IT service management (ITSM) and manufacturing. In ITSM, DSSs can enhance decision-making processes and improve service delivery [38–40]. For manufacturing, DSSs aid in logistics management, production planning, and adapting to changing business environments [41–43]. DSSs have also been used in educational contexts, such as e-learning

environments and instructional systems development, to improve course management and student performance [44–46]. These systems often incorporate advanced technologies like AI, operations research, and semantic technologies to process large datasets and provide valuable insights [40,45].

The latest research has explored decision support systems (DSSs) in various health-care and environmental contexts. In long-term care facilities, DSSs have shown potential to improve medication safety, enhance care delivery, and support clinical decision-making [47–49].

LSTM-based models have even been used to analyze hospital admissions during the pandemic [50]. In IT service management, DSSs can aid in process assessment and improvement [51]. For individuals with long-term conditions, a multidimensional DSS tool has been proposed to support informed healthcare choices [52].

For construction logistics, DSSs have been developed to address urban challenges and support sustainable solutions [53–55]. In data centers, DSS models can optimize waste heat recovery systems [56].

On the environmental side of forest and wildland fire management, DSSs have been developed to enhance suppression resource management [57].

Some papers explore DSSs for SMEs on specific applications, such as servitization [58], credit risk analysis [59], vendor-managed inventory [60], and tax arrear prediction [61]. Others address broader operational challenges, including collaborative web-based operations management [62] and BIM adoption in developing economies [63].

These papers highlight the importance of DSSs in enhancing decision-making capabilities and improving overall business competitiveness for SMEs. Various methodologies are employed, including genetic algorithms, machine learning, and fuzzy synthetic evaluation. The studies reveal that well-designed DSSs can provide significant benefits to SMEs, such as improved performance, cost reduction, and more informed risk assessments [49,64,65].

### 2.3. LSTMs in Financial Times Series Prediction and Generative AI

Long Short-Term Memory (LSTM) networks are a type of recurrent neural network designed to handle long-term dependencies in sequential data [66]. Recent studies have explored LSTM-based decision support systems across various domains.

In oil and gas exploration, an LSTM model with hyperparameter tuning improved log prediction accuracy [67]. For air traffic management, an LSTM-Cubic A\* system enhanced trajectory prediction and threat assessment [68]. In healthcare, LSTM models have shown promise for diabetes diagnosis [69] and cardiovascular disease event prediction [70]. LSTM has also been applied to energy-efficient wireless communication [71] and renewable energy integration in microgrids [72]. In finance, LSTM-based forecasting has been incorporated into deep reinforcement learning systems for automated stock trading [73].

These studies show LSTM's versatility in capturing temporal patterns and improving decision-making across diverse fields, often outperforming traditional methods in prediction accuracy and system performance.

### 2.4. Ethical Considerations and AI

Ethical considerations have become increasingly important as AI systems become more integrated into international business decisions. Issues of transparency, fairness, and privacy have come to the fore [74].

SMEs adopting AI-driven solutions must navigate ethical challenges related to data privacy and algorithmic bias, particularly when operating across different regulatory environments [75,76].

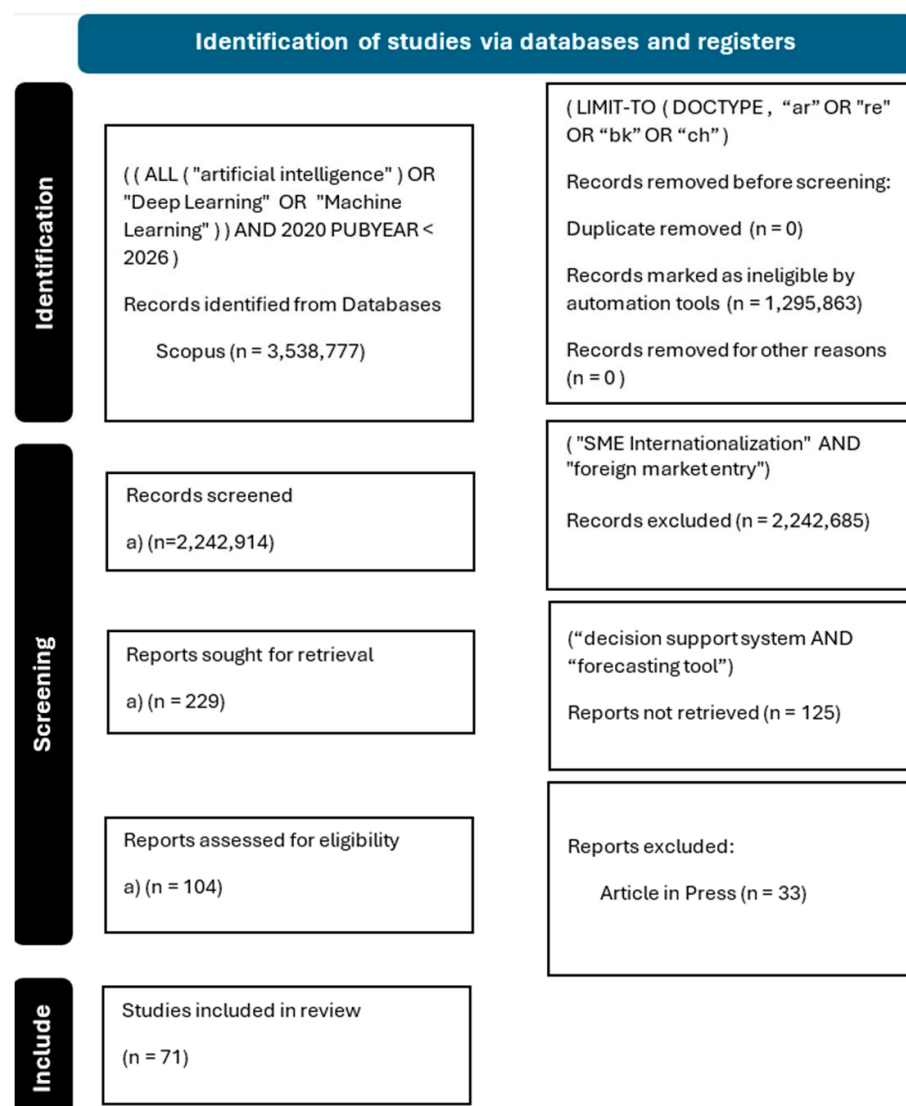
Future research should focus on developing more resource-efficient DL models tailored to the specific needs of internationalizing SMEs. This includes improving interpretability to enhance decision-making transparency, developing transfer learning techniques to mitigate data scarcity issues, and creating AI solutions that adhere to ethical principles while improving decision-making capabilities in international contexts.

### 3. Method

This study employs a mixed methods approach to develop and evaluate a generative AI deep learning recommender system for SME internationalization. The research design follows a sequential exploratory strategy, as described by Creswell & Creswell [77], integrating systematic literature review, qualitative data collection and analysis, and conceptual model development.

#### 3.1. Systematic Literature Review

A comprehensive systematic literature review was conducted following the PRISMA guidelines [78], as revealed in Figure 1. The review aimed to identify key factors influencing SME internationalization and existing AI applications in this domain.



**Figure 1.** PRISMA 2020 statement of the systematic literature review. SLR PRISMA flow—last accessed on 9 May 2025.

The search strategy included Web of Science, Scopus, and IEEE Xplore databases, focusing on peer-reviewed articles published between 2019 and 2025. Search terms combined “artificial intelligence,” or “machine learning,” or “deep learning,” limited to articles and reviews, books, and book chapters, and finally refined by “SME Internationalization” or “foreign market entry,” and “forecasting tool” and then ordered by citations (33 documents were excluded by article in press). From 3,538,777 initial results, 71 studies were selected for in-depth analysis after rigorous screening.

### 3.2. Conceptual Model Development and Validation

A conceptual model for the AI deep learning recommender system was developed using Visual Paradigm Professional 17.2, utilizing SysML v1 diagrams to ensure clarity and precision in system representation. The validation of this conceptual model draws on multiple sources—several prior systematic literature reviews, expert interviews, and the current systematic review of comparable solutions, which includes a detailed comparison of tools and identification of best practices.

For further research validation, a two-round Delphi study with a panel of twelve experts in artificial intelligence, international business, and SME management is proposed, following the methodology proposed by Linstone and Turoff [79]. At present, the model’s parameters are informed by insights from systematic reviews in determinants of SME internationalization success and previous interviews with specialists. This approach is considered appropriate for the current stage, as the platform remains in the conceptualization phase and most variables—approximately ninety-nine percent—are grounded in decades of well-established research. This phased and multi-source validation strategy ensures both theoretical rigor and practical relevance while allowing for additional empirical validation as the project progresses.

Inputs are, therefore, the 184 variables identified through prior work (SLRs and expert interviews). The model’s architecture follows a modular design, with components for feature selection, explainable AI, and user interaction, which will be described in detail in subsequent sections. An internal ERD was also created to ensure the systematic organization of variables and their relationships (currently part of another article under review).

Thus, the development and validation of the DEEPEIA conceptual model followed a multi-stage, evidence-based process, as follows:

(a) Foundational Literature Review (SLR1): The initial variable set was established through a comprehensive systematic literature review (SLR) conducted in 2023, which identified 181 determinants of SME internationalization from international business management literature. This foundational work ensured the model was grounded in established theory and empirical findings.

(b) Review of State-of-the-Art Digital Solutions: A second, targeted SLR was undertaken specifically for this study, focusing on recent and hard-to-find AI-based and digital recommender systems for SME internationalization, including the Greek platform and other comparable international solutions. This review enabled benchmarking and informed the design of DEEPEIA’s architecture and features.

(c) Empirical Validation and Variable Refinement: Empirical validation was achieved through semi-structured interviews with twelve internationalization experts. Each interview session comprised two distinct parts: the first part elicited best practices in SME internationalization (as detailed in the first empirical article, 2024), while the second part explored the features and functionalities considered essential for an AI-driven foreign market recommender system (focus of the second empirical article, currently under review, 2025). Thematic analysis of interview data guided the final selection and categorization of variables and features, which were subsequently integrated into the conceptual model.



The resulting set of 184 variables is categorized and mapped in detail in a forthcoming article, including a complete entity–relationship diagram (ERD) illustrating their interconnections (Calheiros-Lobo et al., under review). This user-driven, empirically validated approach ensures that DEEPEIA addresses real-world SME needs and stands apart from more prescriptive, top–down models.

A formal citation to the companion article will be added as soon as it is published.

#### 4. Findings

Table 1 depicts the top ten results of the systematic review, ordered by citations.

**Table 1.** SLR top 10 documents ordered by citations.

| Authors | Title                                                                                             | Source                                              | Cited |
|---------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------|
| [80]    | Organisational learning, learn. org., and learn. orientation: [...] review and framework          | <i>Human Resource Management Review</i>             | 103   |
| [81]    | Int. social SMEs in emerging countries: Do govts. support their international growth?             | <i>Journal of World Business</i>                    | 75    |
| [82]    | Customer Relationship Mgmt. and its impact on entrepreneurial marketing: a literature review      | <i>Int. Entrepreneurship and Management Journal</i> | 51    |
| [83]    | Examining knowledge transfer and networks: an overview of the last twenty years                   | <i>Journal of Knowledge Management</i>              | 33    |
| [84]    | Digital Capabilities, Their Role in Business Model Innovativeness, and the Int. of SMEs           | <i>IEEE Transactions on Eng. Management</i>         | 30    |
| [85]    | Is offshoring dead? A multidisciplinary review and future directions                              | <i>Journal of International Management</i>          | 30    |
| [86]    | Systematic review of institutional innovation literature: towards a multi-level Mgmt. model       | <i>Management Review Quarterly</i>                  | 28    |
| [87]    | More power for international entrepreneurs: the effect of digital readiness of economies on [...] | <i>Journal of International Entrepreneurship</i>    | 15    |
| [88]    | The determinants of exp. perf. in the digital transformation era: [...] from manuf. firms         | <i>M International Journal of Emerging Markets</i>  | 13    |
| [89]    | Assessing SMEs' internationalisation strategies in action                                         | <i>Applied Sciences (Switzerland)</i>               | 9     |

Authors own elaboration from SCOPUS; last accessed 9 May 2025.

The SLR highlights the growing importance of artificial intelligence (AI) and machine learning in international business and market expansion. Large Language Models have demonstrated remarkable capabilities in natural language processing and other tasks [90]. AI is transforming global business operations, aiding in efficiency and innovation [91]. Predictive analytics and machine learning are being used to guide strategic planning and market expansion [92]. These technologies are helping businesses identify new target markets and develop new tools like DEEPEIA [93] and analyze international trade patterns [94]. For SMEs, AI readiness and business model innovation are crucial for internationalization [95]. Digital platforms are also enabling small firms to enter international markets [96]. In the field of remote sensing, foundation models like SpectralGPT are being developed to handle spectral data [97].

The literature on artificial intelligence in international business reveals several persistent gaps that limit both theoretical advancement and practical application. Despite the proliferation of AI tools, there remains a limited and fragmented understanding of how AI influences international business strategies, operational practices, and overall firm activities. The systematic integration of AI, particularly predictive analytics, and machine learning, into strategic decision-making and market expansion processes is still underdeveloped, underscoring the need for robust frameworks that align these technologies with

international business planning. Research addressing AI readiness and adoption among small and medium-sized enterprises is notably scarce, especially regarding the dynamic interplay between AI capabilities, frugal innovation, and business model transformation as SMEs pursue internationalization. Furthermore, the application of AI across specific international business functions—such as market selection, entry mode choice, foreign exchange management, international human resource management, and cross-cultural operations—remains insufficiently explored. Additional gaps persist in understanding the organizational and workforce adjustments necessary for effective AI integration, the potential of digital platforms and AI to facilitate small firm internationalization, and the use of AI for spectral data analysis in remote sensing, which holds promise for industries reliant on geographical and environmental intelligence. Addressing these gaps is essential for unlocking the full potential of AI to drive innovation and competitive advantage in the global business landscape.

The SLR also suggested that future research in AI and IB should prioritize the development of comprehensive frameworks that integrate AI, particularly machine learning and predictive analytics, into strategic planning for global market expansion. There is a need to investigate how AI can enhance decision-making by providing deeper insights into complex market dynamics and supporting organizations in rapidly evolving environments. Research should also focus on the conditions that enable successful AI adoption among SMEs, examining the interplay between AI readiness, frugal innovation, and business model transformation in the context of internationalization. Further studies are warranted to explore the application of AI across specific IB functions, including market selection, entry modes, foreign exchange management, human resource management, and cross-cultural operations. Empirical work should address how digital platforms and AI can help small firms overcome traditional barriers to internationalization. The development and use of AI models for spectral remote sensing also merits attention, given their potential for industries reliant on geographical and environmental data. Additionally, research should examine the organizational and workforce adjustments required for effective AI integration, the role of AI in enhancing cultural intelligence, and the ethical and regulatory challenges associated with AI in global business. Finally, longitudinal studies are essential to assess the long-term impact of AI adoption on firm strategy, performance, and sustained competitiveness in international markets.

## 5. Discussion

### 5.1. Theoretical Contribution

Following their prior research in SME internationalization, the authors undertook the conceptualization of DEEPEIA, a Generative AI Deep Learning Recommender that identifies the optimal foreign market for any SME product or service.

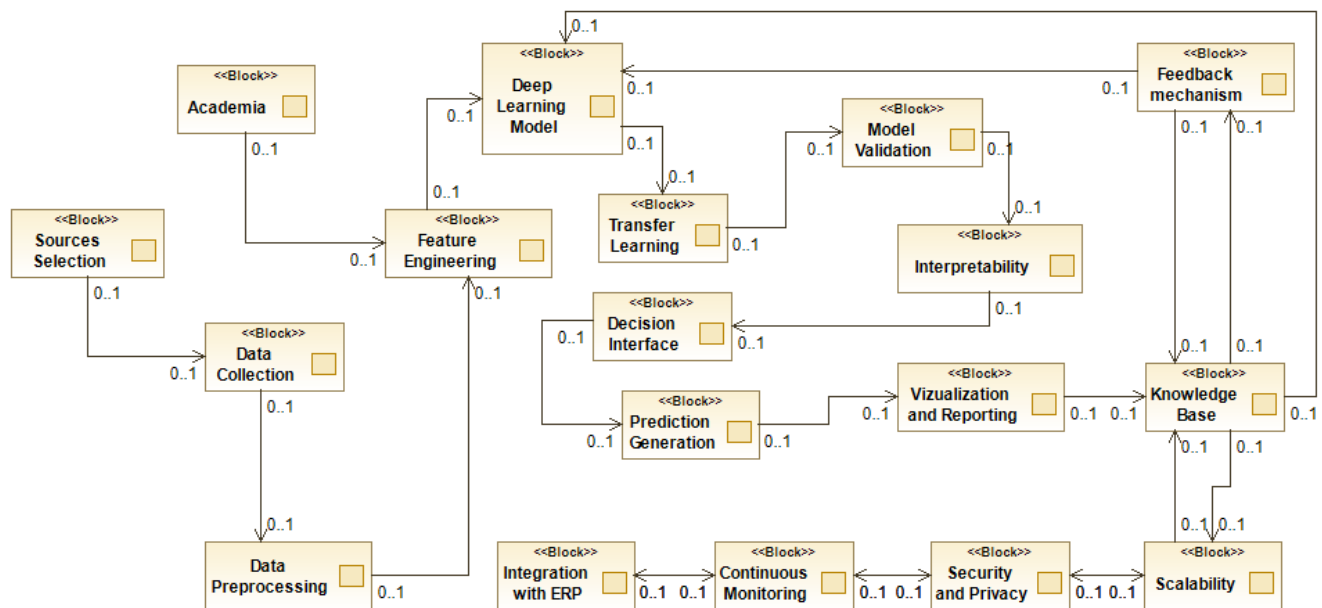
It differs from other solutions for its ability to deal with ETIM item harmonization and the capacity to put the SME owner in contact with the decision-maker for each product or service in the ideal target market.

### 5.2. DEEPEIA—Generative Deep Learning Optimal Foreign Market Recommender

This study conceptualizes a comprehensive framework for developing a machine learning model to analyze and predict market trends, integrating advanced techniques in data collection, feature engineering, model architecture, and explainability.

Figure 2 depicts the initial Block Diagram of DEEPEIA, using Modelio 5.4 software.





**Figure 2.** Initial block diagram of the concept. Authors' own elaboration via Modelio 5.4 software.

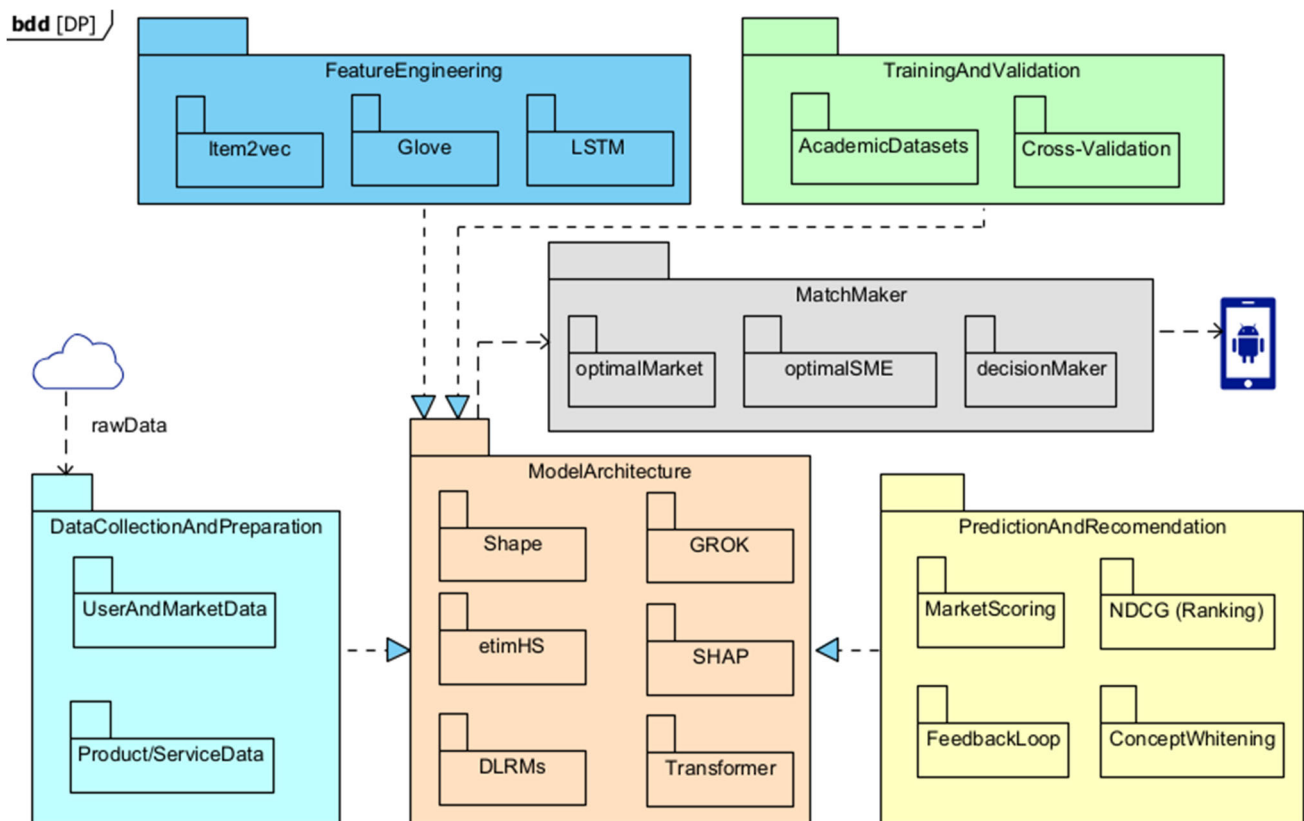
Figure 3 reveals DEEPEIA's SysMLv1 Package diagram, with its MatchMaker innovative tool, where the intelligent decision support system can find not only the ideal market but also the ideal foreign contact for each product and service, conceptualized with Visual Paradigm 17.2 Professional. It also reveals the effort to solve one of the gaps the authors encountered during their research, which is the ability to work with normalization/harmonization at the level of product/service and not categories, as the other solutions, per the authors' knowledge.

DEEPEIA begins with data collection and preparation, leveraging diverse sources such as Comtrade for trade statistics, WITS for tariffs and trade agreements, World Bank Open Data for economic indicators, and the Twitter API for real-time sentiment analysis. Apache Spark 4.0.0 is employed to process this extensive dataset, ensuring integration and cleaning of data from multiple sources into a unified format, like ETIM. This robust preprocessing stage ensures the reliability of the subsequent analytical processes.

In the feature selection and engineering phase, the BORUTA algorithm identifies critical variables from an initial set of 183 features, emphasizing those that enhance predictive accuracy. Feature engineering techniques include Min-Max scaling for the normalization of continuous variables and one-hot encoding for categorical data. Additionally, Retrieval-Augmented Generation (RAG) dynamically enriches features with contextual insights from recent reports or expert analysis, thus enhancing the relevance of variables.

The model architecture employs a hybrid approach combining Long Short-Term Memory (LSTM) layers to capture temporal dynamics and dense layers with ReLU activation for non-linear transformations. A transformer-based attention mechanism further refines variable interactions. To enhance interpretability, a custom concept-specific layer is introduced using Concept Whitening, enabling neurons to detect distinct economic or cultural concepts. The output layer utilizes a softmax function to provide market scoring or ranking.

For training and validation, DEEPEIA optimizes performance using a dual-objective loss function that combines Mean Squared Error (MSE) with penalties for model complexity. The Adam optimizer facilitates adaptive learning rates, while L1 regularization promotes sparsity for simpler models. Training incorporates early stopping with a patience threshold of 10 epochs and employs 5-fold cross-validation to ensure robustness.



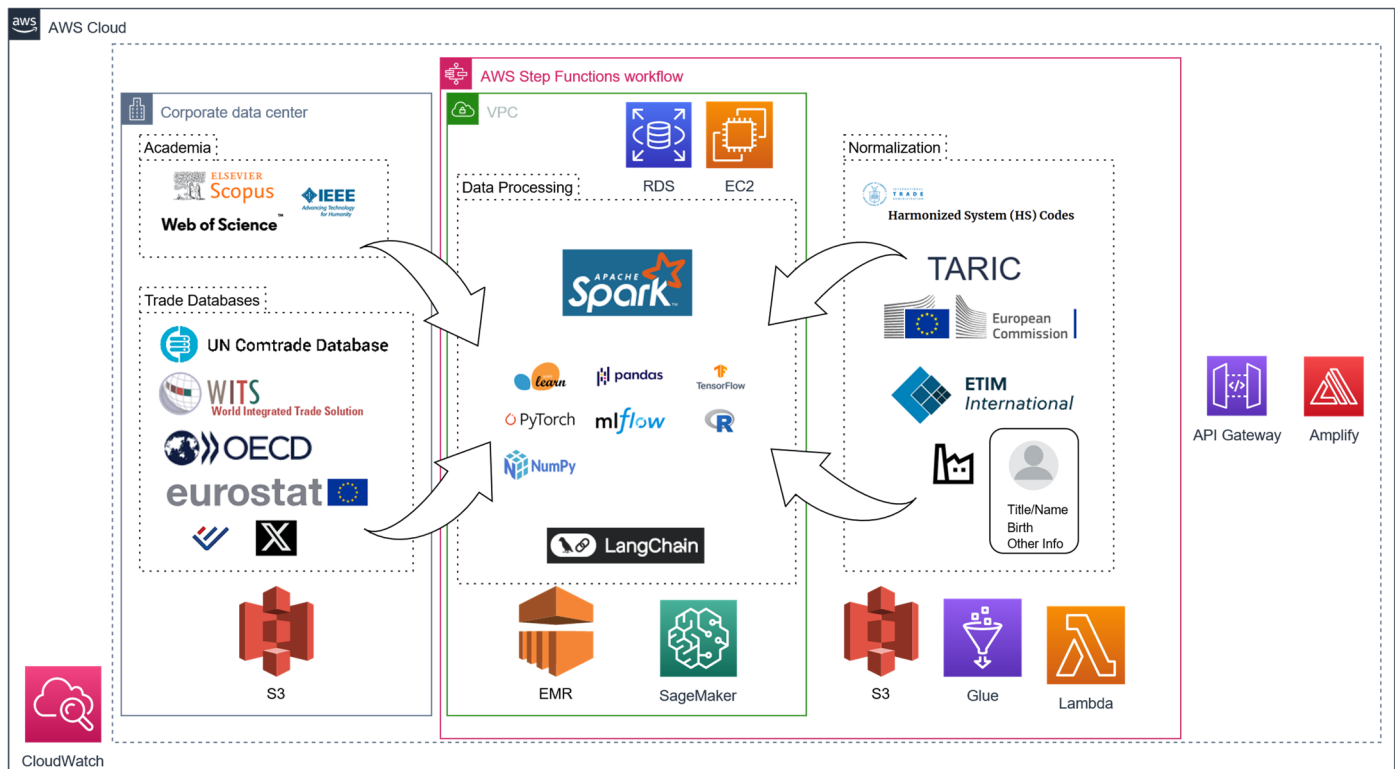
**Figure 3.** DEEPEIA—package diagram with MatchMaker. Authors' own elaboration via Visual Paradigm 17.2 Professional software.

Post-processing focuses on explainability, employing SHAP values to quantify feature contributions to predictions. These insights are translated into natural language explanations using GROK3/GPT-4, enhancing user understanding of recommendations. The user interface integrates React for interactive dashboards and D3.js for visualizations like world maps and charts. Users can interactively adjust variable importance via sliders or query explanations through an API.

Finally, the intelligent decision support system is deployed using TensorFlow Serving on AWS infrastructure, with Kubernetes ensuring scalability. A feedback loop collects user input through forms, which informs periodic model retraining to refine both accuracy and interpretability over time.

This model is an innovative integration of advanced machine learning techniques with practical tools for explainability and user interaction, making it applicable for dynamic market analysis scenarios. Its design aims to balance the complexity needed for accurate export market recommendations with the transparency required for user trust and understanding, leveraging explainability techniques both during and post model processing.

Figure 4 reveals DEEPEIA's AWS Cloud. To implement a cloud-based application capable of processing company data, integrating external databases (e.g., UN Comtrade, WITS, OECD), and generating market recommendations, Amazon Web Services (AWS) provides a scalable infrastructure tailored for such advanced workflows.



**Figure 4.** DEEPEIA—AWS data processing diagram. Authors’ own elaboration via Visual Paradigm 17.2 Professional software.

The process begins with secure data ingestion and storage using Amazon S3, which ensures durability and seamless integration with subsequent analytics services. External data sources are accessed via APIs or scraping mechanisms, facilitated by AWS Lambda’s serverless architecture, enabling efficient event-driven operations with no manual server management.

Once ingested, large-scale datasets are processed using Amazon EMR, which runs Apache Spark in a managed environment optimized for parallel computation. This eliminates the operational overhead associated with deploying and maintaining big data frameworks while ensuring high performance.

For Extract-Transform-Load (ETL) operations and normalization tasks involving classifications like HS or TARIC, AWS Glue provides an automated service that integrates seamlessly with S3 and EMR. These preprocessing steps prepare the data for machine learning workflows.

Machine learning models are developed and deployed using Amazon SageMaker, which supports frameworks such as TensorFlow 2.18.0 and PyTorch 2.7 natively (as of July 2025). SageMaker simplifies the lifecycle of model training, tuning, and deployment while offering scalability to accommodate growing computational demands.

For advanced recommendation generation, LangChain’s Retrieval-Augmented Generation (RAG) techniques can be integrated within SageMaker to dynamically incorporate external knowledge sources into predictive analytics.

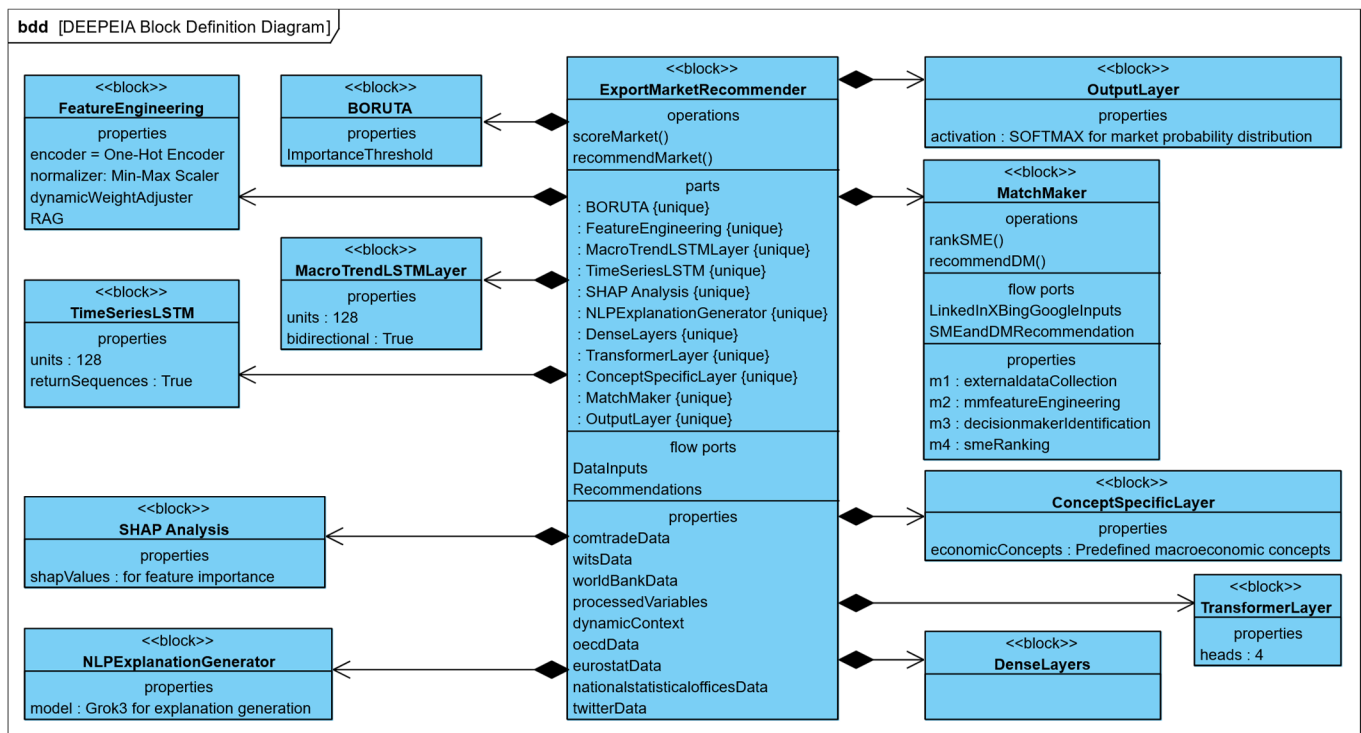
The orchestration of these interconnected processes is managed by AWS Step Functions, which automate the pipeline from data ingestion to output delivery. This ensures that workflows—spanning data retrieval, processing, normalization, model training, and recommendation generation—are executed seamlessly and efficiently.

Monitoring tools like Amazon CloudWatch provide real-time insights into system performance and reliability.

By leveraging AWS’s managed services, this architecture not only streamlines complex workflows but also offers scalability and flexibility essential for modern big data applications.

Integrating innovative techniques, such as RAG within a cloud-native infrastructure, exemplifies how AWS enables organizations to transform vast datasets into actionable insights efficiently and securely.

Figure 5 depicts the SysML v1 Block Definition Diagram (BDD) of DEEPEIA. The BDD defines the hierarchical structure of the foreign market entry recommender system, identifying its main block and constituent parts.

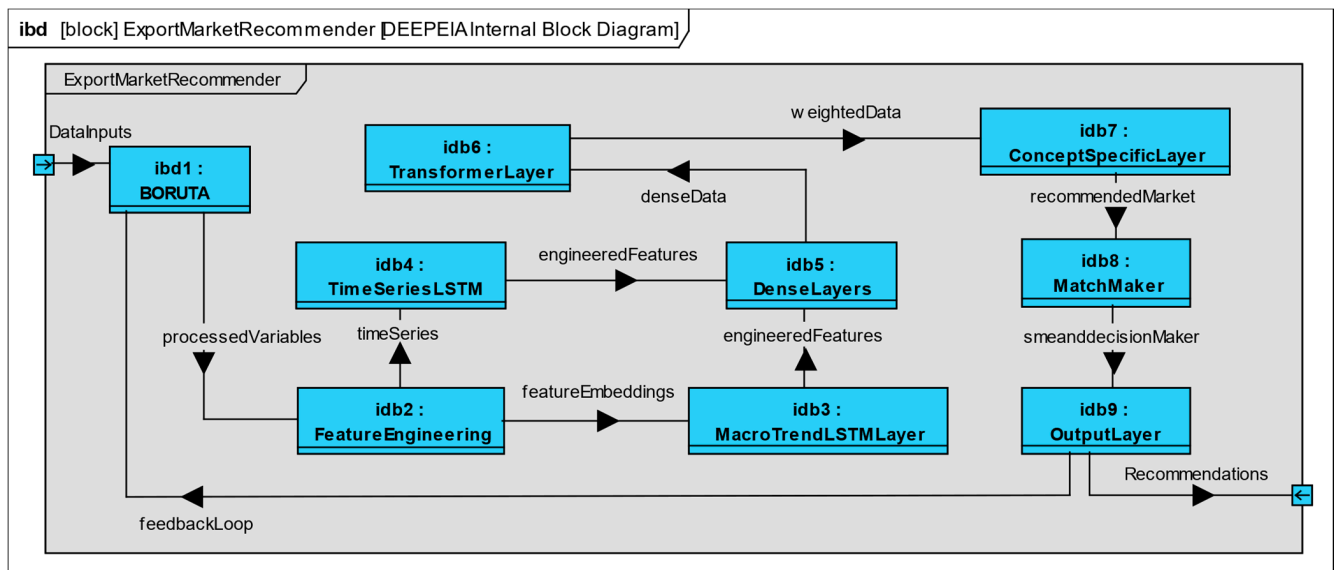


**Figure 5.** DEEPEIA—SysMLv1 block definition diagram. Authors’ own elaboration via Visual Paradigm 17.2 Professional software.

The primary block, **ExportMarketRecommender**, encapsulates nine subcomponents: **BORUTA** (feature selection), **FeatureEngineering** (data refinement and context addition), **MacroTrendLSTMLayer** and **TimeSeriesLSTM** (temporal data processing), **DenseLayers** (non-linear feature processing), **TransformerLayer** (attention mechanisms), **ConceptSpecificLayer** (concept interpretation), **OutputLayer** (market scoring), **SHAPAnalysis** (post hoc explainability, also known as interpretability), and **NLPExplanationGenerator** (natural language explanations).

These subcomponents are connected to the main block via shared composition relationships, denoting their integral role in the system. The modularity captured in the BDD ensures scalability and clarity in understanding component interactions.

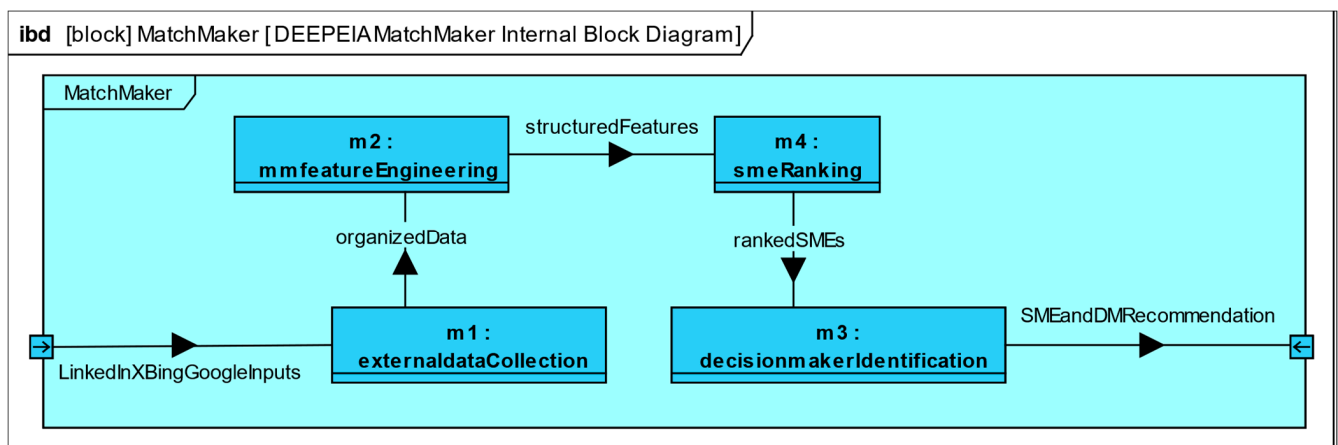
Figure 6 presents DEEPEIA’s **ExportMarketRecommender** SysML v1 Internal Block Diagram (IBD). It highlights the sequential progression of data through its components, as follows: **BORUTA**, **FeatureEngineering**, **TimeSeriesLSTM**, **MacroTrendLSTMLayer**, **DenseLayers**, **TransformerLayer**, **ConceptSpecificLayer**, and **OutputLayer**. Output is processed and returned to **ExportMarketRecommender** for market scoring.



**Figure 6.** DEEPEIA—SysMLv1 ExportMarketRecommender internal block diagram (IBD). Authors' own elaboration via Visual Paradigm 17.2 Professional software.

Following prediction, data flows through SHAPAnalysis and NLPExplanationGenerator to produce interpretable explanations. Control flows originating from ExportMarketRecommender oversee task execution across all subcomponents. Additionally, the ports on ExportMarketRecommender enable external data inputs (In) and outputs (Out), ensuring efficient integration with external systems.

Figure 7 determines the internal block diagram of the MatchMaker part of DEEPEIA, its components, flows, and inputs and outputs.



**Figure 7.** DEEPEIA—SysMLv1 MatchMaker internal block diagram (IBD). Authors' own elaboration via Visual Paradigm 17.2 Professional software.

The MatchMaker AI workflow is a sophisticated system designed to identify and rank SMEs and their decision-makers within the optimal foreign market. The process begins with dynamic data acquisition from external sources like LinkedIn, X (formerly Twitter), and Bing/Google Search APIs. Unstructured data is filtered and organized into datasets categorized by geography or industry in the externaldataCollection phase. This information is then preprocessed to create structured features in the mmfeatureEngineering stage.

In this stage, key steps include cleaning irrelevant data, encoding categorical variables like job titles, and normalizing metrics such as company size or engagement scores. The preprocessed data is integrated with internal insights from earlier stages. Once structured

features are prepared, the workflow ranks SMEs using the smeRanking algorithm. This algorithm evaluates SMEs based on their alignment with predefined business goals through collaborative filtering and similarity scoring techniques. Metrics such as industry relevance and growth potential are considered.

Following SME ranking, the decisionmakerIdentification block identifies key decision-makers within prioritized SMEs. NLP techniques like Named Entity Recognition (NER) extract roles such as CEO or Managing Director from textual data sources like LinkedIn profiles. Sentiment analysis evaluates decision-makers' alignment with business objectives based on their public statements or social media activity. Profiles are validated by cross-referencing multiple sources to ensure accuracy and reliability.

The final stage consolidates outputs into a SMEanDMRecommendation, combining macroeconomic insights with targeted client identification. The output includes a ranked SME alongside its decision-maker within the optimal foreign market for entry.

Together, these three SysML diagrams offer a comprehensive view of the ExportMarketRecommender's architecture and behavior, enabling effective analysis, design, and verification of this complex recommendation system.

Figure 8 demonstrates the frontend of the actual prototype of DEEPEIA, explained as follows.



**Figure 8.** DEEPEIA—UIX. Authors' own elaboration via Visual Paradigm 17.2 Professional software.

## UIX

The User Interface Experience (UIX) for DEEPEIA, focusing on macroeconomic insights, is described as follows:

The authors propose a user-centric UI/UX concept that complements the previously described machine learning framework by providing an intuitive, interactive interface for economic and market analysis. The integration of a sophisticated dashboard layout enhances the accessibility and usability of the underlying predictive model. The dashboard is structured into distinct sections to streamline user interaction. The header prominently features the "DEEPEIA" logo alongside a user profile dropdown for settings or logout options. The main section is divided into input and output areas. The input area includes a product details dialog where users can specify product characteristics, with fields tailored to economic considerations such as price sensitivity and market alignment. An innovative economic focus slider allows users to dynamically adjust the weight of macroeconomic indicators in the recommendations, providing flexibility in aligning outputs with specific business needs.



The output area incorporates advanced visualization tools for actionable insights. An interactive world map, powered by D3.js, highlights countries based on macroeconomic health and recommendation scores, with hover functionality displaying key economic indicators. A ranked list of top recommendations is presented alongside visual elements such as economic-themed gauges or bars to depict scores. Upon selecting a specific market, users gain access to a detailed view that includes an overview of economic indicators (e.g., GDP, inflation, unemployment), a product fit analysis emphasizing economic alignment, and charts illustrating macroeconomic trends and variable impacts using SHAP values for explainability. Additionally, a dedicated explanation section provides natural language insights into the economic factors influencing recommendations, with an interactive query feature enabling users to explore specific trends or indicators.

The implementation's architecture ensures seamless functionality and responsiveness. The frontend is developed using React for dynamic state management and real-time updates of economic data. The backend employs Python 3.13.5 with Django 5.2.4 (LTS) to manage logic, integrate economic data sources, and interface with the deep learning model. Real-time insights are facilitated through RESTful APIs that pull updated economic data, ensuring that recommendations remain current. Deployment leverages AWS infrastructure, including EC2 for processing, S3 for data storage, and RDS for relational data management. Security measures include OAuth 2.0 authentication and HTTPS protocols to safeguard sensitive data.

Accessibility is prioritized through adherence to WCAG 2.1 guidelines, ensuring that visualizations are usable by all users. Features such as variable sliders enable real-time adjustments to recommendations based on user-defined parameters, while export options allow users to generate comprehensive reports in PDF format containing recommendations, analyses, and visualizations.

This UI/UX framework not only democratizes access to complex economic analysis but also bridges the gap between advanced machine learning models and end-user interpretability. By integrating intuitive design elements with robust backend functionality, the system empowers users to make informed decisions grounded in real-time macroeconomic insights. This approach underscores the importance of combining technical sophistication with usability in applied machine learning systems for market analysis. This UIX design aims to make complex economic data readable and actionable, allowing users to leverage macroeconomic insights for better market entry decisions while maintaining the explainability and interactivity of the system.

### 5.3. Managerial and Policy Implications

For SME owners, the implications are substantial. By integrating AI-driven analytics, real-time market sentiment, and comprehensive trade data, such a platform enables SME managers to make more informed, data-driven decisions about international expansion, product positioning, and partner selection. Instead of relying on intuition or generic trade statistics, managers gain access to tailored market recommendations, transparent rationales for each suggestion, and, at the premium level, direct and immediate contact with key market players, streamlining the process of establishing international partnerships. This can significantly reduce the time, cost, and uncertainty traditionally associated with entering new markets while also improving compliance and documentation accuracy through automated, up-to-date regulatory checks.

For policymakers, platforms like DEEPEIA offer new tools for supporting SME internationalization and monitoring trade flows. The granular, real-time data generated can inform the design of targeted export promotion programs, identify emerging market opportunities or risks, and help policymakers understand the evolving needs of the SME

sector. More broadly, the adoption of such AI-powered trade platforms can enhance national competitiveness by fostering a more agile, resilient, and innovation-driven export ecosystem while also highlighting the importance of building regulatory frameworks that keep pace with technological advances in digital trade.

Consultants specializing in SME internationalization are an important audience for DEEPEIA. The platform's ability to identify the optimal foreign market and recommend a suitable foreign champion enhances the internationalization process but does not replace the role of consultants. While AI can sometimes produce inaccurate or hallucinated information, having the right contact is only part of the process. Someone must engage with the foreign champion and provide all the necessary context, allowing SME owners to focus on their core strengths. By supporting market identification and contact discovery, DEEPEIA strengthens the work of consultants and helps SMEs approach international expansion with greater confidence and clarity.

By leveraging the AI-driven recommender system proposed in this study, firms can identify and target their optimal foreign markets with greater precision and even find the best contact (which is usually extremely difficult to do). This approach not only promises to spare financial resources but also to enhance the efficacy of internationalization efforts. It represents a move away from broad, unfocused marketing endeavors, like decades-long tradeshow appearances, with non-existent Return on Investment (ROI), towards data-driven, strategically aligned market entry decisions. Consequently, this research provides a foundation for SMEs to optimize their international expansion strategies, potentially accelerating their global market penetration while minimizing resource wastage.

**Practical Case:** Imagine a Portuguese SME owner who manufactures luxury cotton shirts and wants to identify the best international market for their bestseller. Using a smartphone or laptop, the owner accesses the conceptual DEEPEIA platform, which begins with a simple AI prompt: "How can I help you?" After the user describes the business and objective, DEEPEIA analyzes a comprehensive dataset of 184 variables—including detailed product attributes beyond the standard HS 6105.10 [98] classification for knitted cotton shirts—academic research, trade statistics, and real-time social media sentiment from platforms like X (formerly Twitter). Unlike existing platforms such as OEC or AEC, which primarily provide aggregated trade data without distinguishing product quality or market nuances, DEEPEIA integrates these diverse sources to generate actionable insights tailored to the SME's unique product. The platform's explainable AI identifies Norway as the optimal market, supported by evidence of strong demand for premium textiles, positive consumer sentiment online, and a favorable retail environment. The SME owner can access a basic country report free of charge, which explains why the market is a good fit. However, only paying clients gain access to DEEPEIA's Matchmaker feature, which goes beyond a mere contact list by pinpointing the most relevant "Foreign Champion"—a key agent, importer, or retailer—and enables direct communication via phone, WhatsApp, or other subscribed channels, facilitating immediate and effective engagement. This integrated, data-driven approach empowers SME owners to move beyond generic trade classifications and static data, offering a clear, practical path to international expansion grounded in both quantitative analysis and real-world connections.

#### 5.4. Comparison with World-Renowned Platforms

Table 2 reveals the essential differences between MIT's Observatory of Economic Complexity and Harvard's Atlas of Economic Complexity and the concept of DEEPEIA. The latter was inspired on both and are meant to be complementary. In fact, many of the macro-economic data would be gathered via their APIs and downloadable datasets. The work they have performed until today, in foreign country analysis, is amazing. As

summarized in the table, DEEPEIA's core concept is micro-economical (SMEs), while the other two are macro. Another innovative approach is the disposition to work with more precise harmonization prisms. Forecasting or prediction based on broad categories only work on a macro-level.

**Table 2.** Comparative features between DEEPEIA and major platforms.

| Feature                       | DEEPEIA (This Study)            | MIT Observatory  | Harvard Atlas    |
|-------------------------------|---------------------------------|------------------|------------------|
| Level of Analysis             | ETIM product/service            | Product category | Product category |
| Variable Set                  | 184 (SLR + expert)              | ~60 (trade/econ) | ~50 (trade/econ) |
| Explainable AI                | Yes (SHAP, NLP)                 | No               | No               |
| Decision-Maker Identification | Yes (MatchMaker)                | No               | No               |
| Data Sources                  | Trade, econ, social, e-commerce | Trade, econ      | Trade, econ      |
| User Customization            | High (sliders, explainability)  | Low              | Low              |
| Real-Time Updates             | Yes (planned)                   | No               | No               |
| Cloud Infrastructure          | AWS, scalable                   | Not specified    | Not specified    |

Authors own elaboration.

### 5.5. Future Deep Learning Platform Implications

This study reveals that this type of platform should focus on user-friendliness and the ability to contact the decision-maker.

### 5.6. Recent Similar Work

During this research, a study on a conceptually similar AI-based foreign market recommender was published by a group of researchers in Greece [93]. It is important to note that the present conceptualization of DEEPEIA was developed independently and prior to the appearance of that work. While both models share the overarching goal of leveraging AI for foreign market selection, DEEPEIA distinguishes itself through its comprehensive integration of 184 variables—identified via a rigorous systematic literature review and expert interviews—compared to the more limited variable set in the Greek study. Furthermore, DEEPEIA introduces unique features such as ETIM-level product/service harmonization and the MatchMaker module for decision-maker identification, which are not addressed in the Greek framework. The parallel emergence of these concepts in the literature underscores the timeliness and relevance of AI-driven decision support for SME internationalization, while the independent and more extensive development of DEEPEIA highlights its originality and potential impact.

It is also noteworthy that the conceptual foundation and project for DEEPEIA were formally approved approximately two years prior to the authors' awareness of the Greek article [93]. This timeline confirms that DEEPEIA was developed independently, and any similarities are a result of convergent innovation in response to global research needs. In fact, the Greek team effort validates both the importance and the need for platforms like DEEPEIA.

### 5.7. Limitations and Future Research

This study has the limitation of being a conceptualization only partially assessed in the field with few variables, not all 184, due to its complexity and the resources available, to the authors, at the time of writing.

The development of DEEPEIA has underscored a fundamental reality of innovation; while the generation of novel ideas is essential, their true value is only realized through

effective implementation and commercialization. As this project evolved, it became clear that even the most promising concepts require a capable team, technical resources, and, critically, access to capital to transition from theoretical models to practical, market-ready solutions. This experience echoes the broader innovation literature, which emphasizes that sustainable impact emerges not only from creativity but also from execution and the ability to build viable business models around new technologies [99–101]. For researchers, entrepreneurs, and policymakers alike, bridging the gap between invention and innovation remains a central challenge—and opportunity—in the advancement of AI-driven solutions for SMEs.

In small-data regimes, DEEPEIA could face risks such as overfitting, where models memorize limited training data instead of learning generalizable patterns, leading to poor performance on new cases. Generative models may produce unrealistic or uninformative outputs due to insufficient examples to capture true data distributions. They might also miss rare events or subtle trends, creating blind spots that reduce reliability. Explainability can suffer because predictions based on scarce data are harder to interpret and trust. However, if DEEPEIA incorporates strategies like domain knowledge integration, user feedback loops, and hybrid modeling, it may effectively mitigate these issues. Domain knowledge can guide the model toward meaningful patterns, user feedback can help correct errors and adapt to real-world nuances, and hybrid approaches can combine data-driven insights with expert rules to improve robustness. Together, these methods enhance DEEPEIA's ability to deliver reliable, actionable insights despite limited data availability.

Future researchers could focus on implementing, testing, and validating DEEPEIA. This would involve subscribing to all the major international trade, academic, e-commerce, and statistical databases, such as Comtrade, Scopus, Amazon, and Eurostat. A multidisciplinary team should also be assembled to develop, train, and improve the AI forecasting tool so that, per users' requests and according to their inputs, the intelligent decision support system imports the relevant data and embeds it with the discussed features into a worldwide pre-processed dataset. By leveraging multiple RAG agents and AWS data processing services, the sophisticated solution should then be able to analyze, rank, and propose the optimal foreign market, as well as identify the most suitable foreign champion for each specific product and market.

While the current study focuses on the conceptual development and empirical validation of the DEEPEIA model through a literature review and expert interviews, further work could include pilot implementation using a subset of variables to test the system in real-world settings. Additionally, empirical validation via user studies and expert panels may provide deeper insights into usability and effectiveness. Comparative benchmarking with existing platforms could help position DEEPEIA within the broader ecosystem, while iterative refinement based on feedback from SMEs would ensure the platform remains responsive to evolving user needs. These potential directions represent valuable opportunities for extending and strengthening the platform beyond its current conceptual stage.

An initial focus group could be interviewed to assess real-world results and inform qualitative analysis. The GenAI DSS should then be adapted based on such findings and further refined by integrating ongoing insights from SME owners during daily use.

The authors are available to fully collaborate in such an endeavor.

## 6. Conclusions

The authors advanced a GenAI DL Model, which they called DEEPEIA, therefore satisfying R3 and its O3 objective, which conceptualizes that the most valuable way to solve the eternal resource waste of poorly chosen foreign markets, which have a significant societal and environmental impact. It encompasses not only state-of-the-art from academia

but also the direct feedback of practitioners/experts in the field, namely CEOs of highly profitable companies. This innovation distances itself from others [93], like Harvard's Atlas of Economic complexity [102] or MIT's observatory [103], especially by focusing on product/service ETIM level harmonization, and also by crossing, using explainable AI, open economic data with marketplaces, like Amazon or Tenders, and social networks, like LinkedIn or X, to determine not only the ideal foreign market but also the optimal foreign champion [16], allowing for direct contact and even scheduling. Therefore, it unites the best of both worlds, and it has the potential to be a transforming tool in the way SMEs plan their future.

In summary, this study presents a promising solution to support SMEs in making more profitable foreign market entry decisions. Future research should focus on the empirical implementation of this approach, rigorously evaluating its capacity to simultaneously and accurately identify both the optimal foreign market and the most suitable foreign champion for each SME's products or services, thereby facilitating more effective internationalization. Such advancements would not only validate the current findings but also deliver a practical tool to enhance international SME growth.

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